

Getting Started with Big Data Analytics on OpenStack with Your Own ETL Workflow Process

ETL jobs are very common, with IT organisations handling very large data sets. Executing an ETL workflow on OpenStack infrastructure with just a few clicks is a boon for private cloud operators, compared to setting up Hadoop infrastructure on physical hardware or elsewhere. This article targets IT admins and data analysts who wish to do their workload analysis with Hadoop/Spark on top of an OpenStack private cloud.



When thinking about a leading open source cloud computing platform, OpenStack comes to mind, as it controls large pools of compute, storage and networking resources throughout data centres, and has gained much popularity in the technology market. It introduces new services and features in every release cycle to address critical IT requirements. A very important requirement for any IT organisation is to build a robust platform for performing data analytics with large data sets. Big Data is the latest buzzword in the IT industry. This article focuses on how OpenStack plays a key role in addressing Big Data use cases.

Big Data on OpenStack

Nowadays, data is generated everywhere and its volume is growing exponentially. Data is generated from Web servers, application servers and database servers in the form of user

information, log files and system state information. Apart from this, a huge volume of data is generated from IoT devices like sensors, vehicles and industrial devices. Data generated from the scientific simulation model is also an example of a Big Data source. It is difficult to store this data and perform analytics with traditional software tools. Hadoop, however, can address this issue.

Let me share my use case with you. I have a large volume of data stored in an RDBMS environment. It does not perform well when the data grows bigger. I cannot imagine its performance when it grows even bigger. I do not feel comfortable with adopting the NoSQL culture at this stage, but I need to store and process the bulk of my data in a cost-effective way. I also need to scale the cluster at any time and require a better dashboard to manage all of its components. Hence, I planned to set up a Hadoop cluster on top of OpenStack and create my ETL job environment.